

Response informed POD and parametric matrix emulation for RPA and QRPA

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(Dated: September 21, 2026)

Abstract

hello there

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I. INTRODUCTION

A. Related work

The low-rank approximation used below is standard POD. The relevant projection property follows from the low-rank approximation result of Eckart and Young [1]; the snapshot construction is described by Sirovich [2]. The indefinite metric, partner spectrum and stability conditions used for RPA are discussed by Nakada [3, 4].

Cook et al. introduced parametric matrix models in which trainable matrix equations produce the requested outputs [5]. Jin et al. developed two complementary surrogates for nuclear linear response and applied them to dipole polarizability and beta decay [6]. Here the reduced space is constructed from physical information before the matrix model is fitted. The PMM then learns the projected stability matrix, while the external field remains physically projected.

Nakatsukasa, Inakura and Yabana introduced the finite-amplitude method (FAM) for self-consistent HF-RPA and demonstrated it for deformed ^{20}Ne [7]. Avogadro and Nakatsukasa extended it to quasiparticle response [8], and Nikšić et al. developed the relativistic implementation [9]. The DIRQFAM program and its v2.0.0 release provide the backend used here [10–12]. Our calculation adds response-vector export, common-coordinate reduction and stability-matrix actions to this backend. It does not replace the FAM solver.

[TODO: This is a CODEX generated literature discussion, it should be verified/expanded. The rest of the introduction also needs to be filled in]

II. DRIVEN RESPONSE POD

We begin by defining our dimensional reduction method, which is a variant of standard proper orthogonal decomposition (POD). The mathematical construction and optimality result in this section assume that complete driven response vectors are available at a finite set of parameter and frequency points. The construction here regards the full response QRPA setting, but it also transfers to the finite amplitude method considered later. While the information restricted experiments use different variants of the dimension reduction

method that do not fall under the assumptions of the optimality statement, the motivating and archetypical framework of the POD is the one constructed in this section. In the information restricted cases, the suitability of the reduced space is verified numerically, rather than through the optimality theorem proved in this section.

The reduced space is constructed using POD of the sampled driven response vectors. POD is a standard low-rank approximation method where the leading left singular vectors of a snapshot matrix span the subspace that minimizes the mean squared Euclidean projection error of its columns [1, 2]. For completeness we prove this optimality result in the notation used here because the choice and normalization of the snapshots determine which part of the physical problem is being optimized. In our construction the snapshots are not generic eigenvectors of the RPA/QRPA matrix, but rather vectors generated by the driven response at selected parameter and frequency points. Since the POD space is initially defined using the ordinary Euclidean inner product, it does not automatically preserve the algebraic structure of the RPA/QRPA equations. We therefore impose the required particle-hole partner and metric structure after the POD step.

A. Driven responses and normalized snapshots

We begin by defining the vectors used for the POD. Let D be the full doubled response dimension, so that D is even. For physical parameters μ , the Hermitian stability matrix and the indefinite metric are

$$\mathcal{H}(\mu) = \mathcal{H}(\mu)^\dagger \in \mathbb{C}^{D \times D}, \quad \mathcal{N} = \begin{pmatrix} I_{D/2} & 0 \\ 0 & -I_{D/2} \end{pmatrix}. \quad (1)$$

The RPA/QRPA modes solve $\mathcal{H}(\mu)z_\nu = \omega_\nu \mathcal{N}z_\nu$. For each selected parameter, excitation energy and external field, let

$$[\mathcal{H}(\mu) - \zeta(E)\mathcal{N}]x_a(E; \mu) = -\mathcal{F}_a(\mu), \quad \zeta(E) = E + i\eta, \quad \eta > 0, \quad (2)$$

be the corresponding D -dimensional driven response. Here $\mathcal{F}_a(\mu) \in \mathbb{C}^D$ represents the external field labelled by a . We assume that the shifted system is nonsingular at the selected points. The broadening parameter is the imaginary shift η , while when use a Lorentzeian

full width we set $\Gamma = 2\eta$. The strength associated with the driven response is

$$S_a(E; \mu) = -\frac{1}{\pi} \text{Im}[\mathcal{F}_a(\mu)^\dagger x_a(E; \mu)]. \quad (3)$$

Suppose that the selected driven responses are indexed as $x_1, \dots, x_m$ and are nonzero. We define the normalized vectors by

$$\tilde{x}_j = \frac{x_j}{\|x_j\|_2}, \quad j = 1, \dots, m. \quad (4)$$

The snapshot matrix and response covariance are given by

$$X = [\tilde{x}_1 \ \cdots \ \tilde{x}_m], \quad Q_{\text{res}} = \frac{1}{m} X X^\dagger = \frac{1}{m} \sum_{j=1}^m \tilde{x}_j \tilde{x}_j^\dagger. \quad (5)$$

It is Hermitian and positive semidefinite, and

$$\text{tr}(Q_{\text{res}}) = \frac{1}{m} \sum_{j=1}^m \|\tilde{x}_j\|_2^2 = 1. \quad (6)$$

B. POD optimality

Denote the eigenvalues of Q_{res} by $\lambda_1 \geq \dots \geq \lambda_D \geq 0$, with corresponding Euclidean-orthonormal eigenvectors $v_1, \dots, v_D$. We use k for the initial POD dimension and r for the final dimension. For $1 \leq k \leq D$, define

$$V_k = [v_1 \ \cdots \ v_k], \quad S_{\text{POD}} = \text{span}\{v_1, \dots, v_k\}, \quad \Pi_k = V_k V_k^\dagger. \quad (7)$$

Thus Π_k is the Euclidean orthogonal projector onto S_{POD} .

Proposition 1 (POD optimality). *Among all k -dimensional subspaces of $\mathbb{C}^D$, S_{POD} minimizes the average squared Euclidean projection error of the normalized driven responses. Explicitly,*

$$\begin{aligned} \frac{1}{m} \sum_{j=1}^m \|\tilde{x}_j - \Pi_k \tilde{x}_j\|_2^2 &= \min_{\dim S=k} \frac{1}{m} \sum_{j=1}^m \|\tilde{x}_j - \Pi_S \tilde{x}_j\|_2^2 \\ &= 1 - \sum_{\ell=1}^k \lambda_\ell = \sum_{\ell=k+1}^D \lambda_\ell, \end{aligned} \quad (8)$$

where Π_S denotes the Euclidean orthogonal projector onto S . If $k < D$ and $\lambda_k > \lambda_{k+1}$, the minimizing subspace is unique.

Proof. For any k -dimensional $S \subset \mathbb{C}^D$, the Pythagorean identity gives

$$\|\tilde{x}_j - \Pi_S \tilde{x}_j\|_2^2 = \|\tilde{x}_j\|_2^2 - \|\Pi_S \tilde{x}_j\|_2^2. \quad (9)$$

Since $\|\tilde{x}_j\|_2 = 1$, averaging gives

$$\frac{1}{m} \sum_{j=1}^m \|\tilde{x}_j - \Pi_S \tilde{x}_j\|_2^2 = 1 - \frac{1}{m} \sum_{j=1}^m \|\Pi_S \tilde{x}_j\|_2^2. \quad (10)$$

By the properties of Π_S , we have that

$$\|\Pi_S \tilde{x}_j\|_2^2 = \tilde{x}_j^\dagger \Pi_S \tilde{x}_j, \quad \frac{1}{m} \sum_{j=1}^m \|\Pi_S \tilde{x}_j\|_2^2 = \text{tr}(\Pi_S Q_{\text{res}}). \quad (11)$$

Hence minimizing the projection error is equivalent to maximizing $\text{tr}(\Pi_S Q_{\text{res}})$. By using the spectral decomposition of Q_{res} , we get that

$$\text{tr}(\Pi_S Q_{\text{res}}) = \sum_{\ell=1}^D \lambda_\ell \|\Pi_S v_\ell\|_2^2. \quad (12)$$

Since $0 \leq \|\Pi_S v_\ell\|_2^2 \leq 1$ and Π_S has rank k ,

$$\sum_{\ell=1}^D \|\Pi_S v_\ell\|_2^2 = k. \quad (13)$$

By the decreasing ordering of the eigenvalues, the largest possible weighted sum is obtained by assigning unit weight to the first k eigenvectors. Thus

$$\max_{\dim S=k} \text{tr}(\Pi_S Q_{\text{res}}) = \sum_{\ell=1}^k \lambda_\ell, \quad (14)$$

with equality when $S = S_{\text{POD}}$. Substitution gives Eq. (8). When $\lambda_k > \lambda_{k+1}$, equality requires $\|\Pi_S v_\ell\|_2^2 = 1$ for $\ell \leq k$ and zero for $\ell > k$, so the minimizing subspace is unique. $\square$

This proposition is the standard POD optimality result, and means that the leading k eigenvectors of Q_{res} identify the k -dimensional subspace that preserves as much of the selected collection of normalized driven responses as possible with respect to the projection norm. Its significance in our setting lies in the choice of snapshot vectors. The optimized space represents the sampled normalized responses rather than a generic collection of QRPA matrix eigenvectors.

It is important to distinguish the quantity optimized by the POD construction from the quantities used to evaluate our methods later. The optimality result above concerns only the sampled, normalized driven response vectors used to construct the POD space. It does not by itself guarantee that the same space accurately represents driven responses at parameter and frequency points that were not included in the construction, nor does it guarantee accurate strength functions at such withheld points.

The numerical experiments below therefore test a stronger property than the POD optimality statement itself, namely whether a space selected to optimally represent the sampled driven responses also captures the response structure needed to reproduce withheld strength functions. We find this behavior across all of the numerical studies considered here, including the different information access regimes introduced below. As mentioned earlier the precise POD basis construction and available response information differ between these regimes, so the POD optimality statement applies directly only to the complete-response construction, which further illustrates that the accurate reproduction of withheld response structure is a nontrivial empirical result.

C. Partner closure and the restricted metric

We now impose the particle-hole partner structure on the reduced Euclidean subspace. For $X, Y \in \mathbb{C}^{D/2}$, the partner map is given by

$$\mathcal{P} \begin{pmatrix} X \\ Y \end{pmatrix} = \begin{pmatrix} Y^* \\ X^* \end{pmatrix}. \quad (15)$$

It is antilinear, satisfies $\mathcal{P}^2 = I$, and preserves the Euclidean norm. The RPA/QRPA stability matrix commutes with the partner map, $\mathcal{H}\mathcal{P} = \mathcal{P}\mathcal{H}$, while the metric changes its sign, $\mathcal{N}\mathcal{P} = -\mathcal{P}\mathcal{N}$ [3]. We say that a subspace $S \subset \mathbb{C}^D$ is partner closed when $x \in S$ implies $\mathcal{P}x \in S$. Since $\mathcal{P}^2 = I$, this is equivalent to $\mathcal{P}S = S$.

The structural completion used here is the enlarged space

$$S_{\text{cl}} = S_{\text{POD}} + \mathcal{P}S_{\text{POD}} = \text{Ran}[V_k \ \mathcal{P}V_k]. \quad (16)$$

When specified symmetry or field directions are required in the physical model, their partner closed span is also included in S_{cl} .

It remains to ensure that the retained space preserves the indefinite metric $\mathcal{N}$. It is useful to note that

$$(\mathcal{P}u)^\dagger \mathcal{N}(\mathcal{P}v) = -(u^\dagger \mathcal{N}v)^*, \quad (\mathcal{P}v)^\dagger \mathcal{N}(\mathcal{P}v) = -v^\dagger \mathcal{N}v. \quad (17)$$

Consequently, a direction with positive metric norm has a partner with negative metric norm, as well as the converse. Let V_{cl} be a Euclidean-orthonormal basis of the enlarged space. The metric restricted to S_{cl} is represented by

$$G = V_{\text{cl}}^\dagger \mathcal{N} V_{\text{cl}}. \quad (18)$$

Since $\mathcal{N}$ is Hermitian, G is Hermitian and has real eigenvalues. Let

$$Gw_j = \gamma_j w_j.$$

The corresponding full-space directions are $V_{\text{cl}}w_j$. We omit those associated with zero or numerically near zero eigenvalues and define the retained space by

$$S_{\text{ret}} = \text{span} \{V_{\text{cl}}w_j : |\gamma_j| > \tau_{\text{met}}\},$$

where $\tau_{\text{met}} > 0$ is a chosen tolerance. The metric restricted to S_{ret} has no zero eigenvalues and is therefore nondegenerate. The following result shows that this restricted metric has balanced signature, meaning equally many positive and negative eigenvalues.

Theorem 2 (Balanced metric signature). *Let S_{ret} be the partner closed space remaining after the zero restricted metric eigenvalues have been removed. Then the restricted metric has equally many positive and negative eigenvalues. Consequently, $\dim S_{\text{ret}} = 2q$ for some q , and its signature is (q, q) .*

Proof. By construction, the metric is nondegenerate on S_{ret} . Let n_+ denote the number of positive eigenvalues and n_- the number of negative eigenvalues of its restricted metric. These are then the largest dimensions of the positive and negative definite subspaces, respectively. As a consequence of Eq. (17), if $S_+ \subset S_{\text{ret}}$ is a positive definite subspace, then $\mathcal{P}S_+$ is a negative definite subspace. Since $\mathcal{P}^2 = I$, the map is bijective between S_+ and $\mathcal{P}S_+$, hence it preserves their complex dimension. Choose S_+ to have the largest possible positive-definite dimension, namely n_+ . Its partner is a negative definite subspace of dimension n_+ , meaning

$n_- \geq n_+$. By symmetry we also get that $n_+ \geq n_-$, meaning $n_+ = n_-$. Since no zero metric eigenvalues remain,

$$\dim S_{\text{ret}} = n_+ + n_- = 2q. \quad (19)$$

The restricted metric therefore has balanced signature (q, q) . $\square$

D. Metric canonicalization

We now provide the canonical construction of the reduced space and metric. Assume now that the retained partner closed subspace has dimension $r = 2q$ and that its restricted metric is nondegenerate. Let V_{ret} be a Euclidean orthonormal basis of this space and define

$$G_{\text{ret}} = V_{\text{ret}}^\dagger \mathcal{N} V_{\text{ret}}.$$

Since G_{ret} is Hermitian, it can be diagonalized as

$$G_{\text{ret}} = W \Lambda W^\dagger,$$

where W is unitary and

$$\Lambda = \text{diag}(\gamma_1, \dots, \gamma_r)$$

contains q positive and q negative eigenvalues. Define

$$|\Lambda|^{-1/2} = \text{diag}(|\gamma_1|^{-1/2}, \dots, |\gamma_r|^{-1/2}).$$

After ordering the positive eigenvalues first and the negative eigenvalues second, define

$$U_r = V_{\text{ret}} W |\Lambda|^{-1/2}.$$

Then

$$\begin{aligned} U_r^\dagger \mathcal{N} U_r &= |\Lambda|^{-1/2} W^\dagger V_{\text{ret}}^\dagger \mathcal{N} V_{\text{ret}} W |\Lambda|^{-1/2} \\ &= |\Lambda|^{-1/2} \Lambda |\Lambda|^{-1/2} \\ &= \begin{pmatrix} I_q & 0 \\ 0 & -I_q \end{pmatrix}. \end{aligned}$$

We therefore define the canonical reduced metric by

$$U_r^\dagger \mathcal{N} U_r = J_r, \quad J_r = \text{diag}(I_q, -I_q).$$

Importantly,

$$\text{Ran}(U_r) = \text{Ran}(V_{\text{ret}}) = S_{\text{ret}}.$$

Metric canonicalization therefore changes only the coordinates used inside the retained space and does not change the space itself. It should be noted that if partner closure or metric filtering has changed the original POD space, the POD optimality statement applies to the original POD space and not automatically to S_{ret} .

[TODO: Most of the text after this section has been generated by Codex based on my code/results, my own notes and clear instructions about the structure. It should be verified and rewritten. It is there now to give the general idea of the structure and to show the important results.]

E. Numerical basis construction

The schematic calculations first take the SVD of the supplied snapshot matrix. They retain singular directions with $\sigma_j^2 > 10^{-12}\sigma_1^2$. Required directions are partner closed separately, then placed before the retained POD directions and their partners. An SVD removes dependent columns at relative singular-value threshold 10^{-10} . Restricted-metric eigenvalues above 10^{-8} and below -10^{-8} determine the maximum admissible paired dimension; their positive eigenvectors also supply fallback seeds.

For each requested even dimension r , seeds are processed in the order: required directions, POD directions, their partners, and positive-metric fallback directions. Given previously accepted positive vectors u_i , a seed v is orthogonalized as

$$v \leftarrow v - \sum_i u_i(u_i^\dagger \mathcal{N}v) + \sum_i \mathcal{P}u_i[(\mathcal{P}u_i)^\dagger \mathcal{N}v]. \quad (20)$$

The projections are applied successively. A seed is skipped if $|v^\dagger \mathcal{N}v|/\|v\|_2^2 \leq 10^{-8}$; a negative-signature seed is replaced by its partner. Positive seeds with metric norm above 10^{-10} are normalized and retained until $r/2$ are available. The positive block is corrected with the inverse square root of its metric Gram matrix, and its partners supply the negative block. Thus the candidate dimension is chosen after structural processing, not by identifying the first r POD vectors with the final basis. The required vectors and rank-selection criteria differ between Secs. IV and V.

For FAM, POD is computed from the small snapshot Gram matrix rather than a full-space covariance or tall SVD. For a Gram spectrum λ_j , the retained eigenvalues satisfy

$$\lambda_j > \max\{\tau^2 \lambda_{\max}, 8\epsilon_{\text{mach}} \lambda_{\max}, 8 \max(-\lambda_{\min}, 0)\}, \quad (21)$$

where ϵ_{mach} is double-precision machine epsilon and $\tau = 10^{-10}$ for both POD and partner-span orthogonalization. Ordered Cholesky reorthogonalization repairs the resulting POD vectors. The metric eigenvalue threshold is $10^{-7} \max(1, \|G\|_2)$, the near-null ratio threshold is 10^{-8} , and accepted positive seeds must have metric norm above 10^{-7} . A Cholesky normalization of the positive metric Gram matrix preserves the nesting of candidate spaces. Metric, partner and required-direction defects are checked at 10^{-8} . The large matrices are accessed in blocks; no $D \times D$ covariance or metric array is needed.

F. Reduced response

The final basis defines the small stability matrix and projected physical field,

$$\mathcal{H}_r(\mu) = U_r^\dagger \mathcal{H}(\mu) U_r, \quad \mathcal{F}_{r,a}(\mu) = U_r^\dagger \mathcal{F}_a(\mu). \quad (22)$$

The reduced response is obtained from

$$[\mathcal{H}_r(\mu) - \zeta(E)J_r]y_a(E; \mu) = -\mathcal{F}_{r,a}(\mu), \quad x_{r,a}(E; \mu) = U_r y_a(E; \mu). \quad (23)$$

The reduced strength is computed using the same contraction as in Eq. (3),

$$S_{r,a}(E; \mu) = -\frac{1}{\pi} \text{Im}[\mathcal{F}_{r,a}(\mu)^\dagger y_a(E; \mu)]. \quad (24)$$

This is a Galerkin reduction: the full residual $-\mathcal{F}_a - [\mathcal{H} - \zeta\mathcal{N}]x_{r,a}$ is Euclidean-orthogonal to the columns of U_r . In numerical calculations the measured projected metric $U_r^\dagger \mathcal{N} U_r$ is retained rather than overwritten by the ideal J_r . The parameter-dependent transported metric in the FAM calculation is checked separately in Sec. VI. In general, the Galerkin solution is not the Euclidean projection $\Pi_{S_{\text{ret}}} x_a$, so the POD projection theorem is not itself a reduced-response error bound.

The same projected pencil underlies all three settings below. What changes is the information used to construct the space, how the projected matrices are obtained, and whether the reported strength is computed from the shifted equation or the positive-frequency modes. The action-generated vectors are introduced in Sec. V, and the common-coordinate transport and matrix-free projection are introduced in Sec. VI.

III. STRUCTURED REDUCED-OPERATOR PMMS

In this section we define the matrix model used after compression. The purpose is to evaluate the reduced operator at new physical parameters without constructing and projecting the full stability matrix again. PMMs use parameter-dependent matrix equations to generate their outputs [5]. We use a polynomial matrix model for the already projected stability matrix.

A. Projected training matrices

Once U_r has been fixed, training parameters μ_j provide matrices $\mathcal{H}_r(\mu_j)$ through Eq. (22). Let $\phi_\ell(\mu)$ be real polynomial functions of the physical parameters. The fitted polynomial part is

$$\mathcal{H}_r^{\text{fit}}(\mu) = \sum_{\ell=1}^{n_\phi} \phi_\ell(\mu) M_\ell. \quad (25)$$

Each parameter coordinate is scaled by the midpoint and half-span of the selected training points: $\xi = (\mu - c)/s$, with $c = (\mu_{\text{max}} + \mu_{\text{min}})/2$ and $s = (\mu_{\text{max}} - \mu_{\text{min}})/2$. The features are Chebyshev polynomials $T_\ell(\xi)$. For one parameter, $n_\phi = p + 1$ for degree p . For two parameters, the features are $T_i(\xi_g)T_j(\xi_f)$ with $i + j \leq p$, giving $n_\phi = (p + 1)(p + 2)/2$.

The coefficient arrays solve the unweighted matrix least-squares problem

$$\min_{\{M_\ell\}} \sum_j \left\| \mathcal{H}_r(\mu_j) - \sum_\ell \phi_\ell(\mu_j) M_\ell \right\|_F^2. \quad (26)$$

The implementation flattens the matrix entries and uses `numpy.linalg.lstsq` with its default rank cutoff; no ridge penalty or response weighting is added. Underdetermined degrees are excluded, as are rank-deficient two-parameter designs. Coefficients are then projected onto the Hermitian, partner-symmetric subspace. Because the features are real, this linear projection is consistent with the shared entrywise least-squares fit. Validation selects the polynomial degree and, where specified, the training size. The action-based learning curve retains every prescribed training size. Candidate sets and selection rules are specified with each experiment.

Each coefficient matrix satisfies

$$M_\ell^\dagger = M_\ell, \quad \Sigma_r \overline{M_\ell} \Sigma_r = M_\ell. \quad (27)$$

The projection takes the Hermitian part and averages it with $\Sigma_r \overline{M} \Sigma_r$. Thus the polynomial part retains Hermiticity and reduced partner symmetry, but these constraints alone do not guarantee stability.

In paired HFB calculations the known particle-number field gives the reduced null direction $n_r = J_r \mathcal{F}_{r,N}$, using the measured reduced metric in numerical evaluation. Its partner-closed span is orthonormalized in the Euclidean norm, yielding a matrix Z ; dependent directions are removed at relative singular-value threshold 10^{-11} . Set $P = I - ZZ^\dagger$, and let Q be any Euclidean-orthonormal basis of its range. The evaluated operator is

$$\begin{aligned} A &= P \mathcal{H}_r^{\text{fit}} P, \\ \delta &= \max\{0, m - \lambda_{\min}(Q^\dagger A Q)\}, \\ \hat{\mathcal{H}}_r &= A + \delta P. \end{aligned} \tag{28}$$

Here m is the minimum permitted stability eigenvalue on the physical complement. Symmetry averaging and projection are repeated to remove roundoff. The HFB runs use $m = 0$, so the number direction remains null and the complementary stability matrix is nonnegative. If $\|n_r\|_2 \leq 10^{-14}$, the implementation instead uses $P = I$ and no null constraint. In the unpaired action-based RPA calculations $P = I$ and $m = 10^{-10}$; the unpaired FAM calculation uses $P = I$ and $m = 0$. The recorded stability shift is the scalar δ , not a shift applied to the protected null direction. This evaluation step is separate from polynomial fitting and is not a claim that partner symmetry or the null constraint alone ensures stability.

The physical field remains $\mathcal{F}_{r,a}(\mu) = U_r^\dagger \mathcal{F}_a(\mu)$. The emulated response follows from Eq. (23) with $\mathcal{H}_r$ replaced by $\hat{\mathcal{H}}_r$. The PMM does not change the chosen reduced space.

For a complex partner-symmetric Hermitian matrix of dimension $r = 2q$, the blocks contain $q^2 + q(q+1) = q(2q+1)$ independent real entries. Therefore the polynomial part has

$$n_\phi \frac{r(r+1)}{2} \tag{29}$$

independent real coefficients. This is the capacity of the generic complex structured matrix class, not a count of nonzero fitted entries. Real matrices admit additional constraints that are not included in this count. We also distinguish it from the $n_\phi r^2$ entries used when full coefficient arrays are stored.

B. Errors and checks

The full RPA/QRPA response is the target, not the exact many-particle response. Let S be the full response, S_r the response from the exact projection, and $\hat{S}_r$ the response from the learned projected matrix. On a declared energy grid, define

$$\varepsilon(S_1, S_0) = \frac{[\sum_j |S_1(E_j) - S_0(E_j)|^2]^{1/2}}{[\sum_j |S_0(E_j)|^2]^{1/2}}, \quad (30)$$

for a nonzero reference response. We report the three errors separately,

$$\varepsilon_{\text{comp}} = \varepsilon(S_r, S), \quad \varepsilon_{\text{PMM}} = \varepsilon(\hat{S}_r, S_r), \quad \varepsilon_{\text{total}} = \varepsilon(\hat{S}_r, S). \quad (31)$$

Means and maxima of Eq. (30) are taken over the stated test parameters and fields. These averages describe the parameter–field test sets rather than stochastic training replicates. The FAM rank and training-size selections additionally use a maximum absolute curve discrepancy normalized by the peak reference strength, defined in Sec. VID; it is distinct from the maximum of Eq. (30).

For the explicit RPA/QRPA benchmarks, the scored strength is the positive-frequency Lorentzian sum

$$S_a^{(+)}(E) = \sum_{\nu>0} \frac{\Gamma}{2\pi} \frac{B_{\nu a}}{(E - \omega_\nu)^2 + (\Gamma/2)^2}, \quad B_{\nu a} = |z_\nu^\dagger \mathcal{F}_a|^2. \quad (32)$$

The symbol S in the error definitions denotes this curve for the schematic experiments and Eq. (3) for FAM. Neither comparison mixes the two conventions. The schematic moments are spectral sums,

$$m_k = \sum_{\nu>0} \omega_\nu^k B_{\nu a}, \quad k = -1, 0, 1, \quad (33)$$

where physical modes have $\text{Re } \omega_\nu > 10^{-8}$, $|\text{Im } \omega_\nu| < 10^{-7}$ and positive metric norm. Eigenvectors are normalized by $z_\nu^\dagger \mathcal{N} z_\nu = 1$; the numerical positive-mode filter requires the resulting norm to exceed 0.5. Same-sign degenerate multiplets are metric orthonormalized before computing $B_{\nu a}$. In paired HFB cases, the two modes of smallest $|\omega_\nu|$ are removed when the number-field norm exceeds 10^{-12} . All three spectral moments exclude frequencies at or below 10^{-8} , not only the inverse-energy-weighted moment. For a moment m_k , the reported relative error is $|m_k^{\text{model}} - m_k^{\text{reference}}|/|m_k^{\text{reference}}|$, with a denominator floor of 10^{-15} in the implementation. The finite-grid FAM moments are specified in Sec. VID.

The structural checks concern Hermiticity, partner closure, the identity $U_r^\dagger \mathcal{N} U_r = J_r$, opposite-frequency pairing, and imaginary parts of physical eigenfrequencies. In the paired calculations, the particle-number null residual and the displacement of the corresponding zero-frequency pair are recorded separately. The smallest stability eigenvalue on the physical complement is checked in addition to these algebraic defects.

Construction data form the basis and the training matrices. Validation data choose the dimension and allowed PMM settings. Test responses are used only after these choices have been fixed. The matrix-action experiment additionally excludes complete RPA solutions from construction and validation.

IV. COMPLETE-RESPONSE HFB-QRPA BENCHMARK

This section tests the full-response construction when pairing and particle-hole mean fields are treated self-consistently. The full operators and driven-response vectors are available during construction. We compare reduced spaces and PMMs against the full QRPA response, with separate normal and paired intervals.

A. Pairing plus particle-hole model

The model has L doubly degenerate levels and N_p particles. Let $c_{p\sigma}^\dagger$ create a particle in level p and spin state $\sigma \in \{+, -\}$, and define $P_p^\dagger = c_{p+}^\dagger c_{p-}^\dagger$. The many-particle Hamiltonian is

$$\begin{aligned} \hat{H}_{\text{mb}}(g, f) = & \sum_{p,\sigma} \epsilon_p c_{p\sigma}^\dagger c_{p\sigma} - \frac{g}{2} \sum_{p,q} P_p^\dagger P_q \\ & - \frac{f}{2} \sum_{p,q,s} (P_p^\dagger c_{q-} c_{s+} + \text{h.c.}), \quad \epsilon_p = d(p-1). \end{aligned} \quad (34)$$

Here $d = 1$ is the one-particle level spacing and the energy unit, g is the pairing strength, and f multiplies the additional particle-hole mixing term. All three sums over levels include coincident indices. The many-particle Hamiltonian is distinct from the response stability matrix $\mathcal{H}$. The numerical input specifies g and f/g ; the interaction always uses the physical strength $f = g(f/g)$.

The main QRPA calculation uses $L = 7$ and $N_p = 6$. At each parameter, the HFB reference treats both interaction terms self-consistently. The QRPA stability matrix is the constrained

HFB energy Hessian in two-quasiparticle coordinates. Its doubled dimension is

$$D = 2 \binom{2L}{2} = 182. \quad (35)$$

The paired references have a particle-number symmetry direction with zero frequency. The basis construction is augmented with particle-number directions, and the physical field and number direction are evaluated from the HFB reference rather than learned. Their representation is assessed through the reported number-null and frequency diagnostics.

The reference solver compares normal and paired stationary branches, retaining the paired branch when its energy is lower by more than 10^{-9} . A paired iteration starts from the normal reference with pairing-field amplitude $\max(0.05, 0.20|g|)$. Its generalized-density update is 0.75 times the preceding density plus 0.25 times the new density, with at most 3000 iterations and a density-change tolerance of 3×10^{-12} . The accepted paired reference has normalized HFB commutator defect at most 10^{-9} and idempotency and particle-number defects at most 3×10^{-10} . If the paired iteration does not converge, the solver retains the normal branch. The computed phase is labelled paired when the Frobenius norm of the pairing tensor exceeds 10^{-7} . The QRPA Hessian is assembled from the exact linearized mean fields; finite differences are used only for the independent curvature check below.

B. Full-order check and parameter sets

Before compression, a four-level, four-particle calculation checks the full-order construction at $(g, f/g) = (1.2, 0)$ and $(1.2, 0.5)$, corresponding to physical strengths $(g, f) = (1.2, 0)$ and $(1.2, 0.6)$. A directional finite-difference Hessian check uses step 3×10^{-4} . The relative curvature errors are 2.59×10^{-8} and 2.43×10^{-8} , respectively. The stationarity defects are below 9.43×10^{-13} , the particle-number null residuals are below 2.94×10^{-13} , and neither calculation contains complex physical modes.

The three main cases and their separate construction, validation and test sets are listed in Table I. Construction couplings are equally spaced including both endpoints, with validation couplings at adjacent midpoints. The test couplings, not used to construct or select a basis,

are

weak normal: $\{0.2137, 0.2671, 0.3419, 0.4173, 0.4931, 0.5817, 0.6579\}$,

weak paired: $\{0.8371, 0.9293, 1.0791, 1.2217, 1.3741, 1.5269, 1.6897, 1.9471\}$,

strong paired: $\{0.4373, 0.6131, 0.7919, 0.9827, 1.1741, 1.3867, 1.6213, 1.8839\}$.

The weak-normal and strong-paired constructions use five spin-scalar one-body fields. In level space, the nearest-neighbour matrix has entries one at $|p - q| = 1$ and zero otherwise; the range-two matrix has entries $1/|p - q|$ for $1 \leq |p - q| \leq 2$ and zero otherwise. The remaining three matrices are symmetrized real Gaussian matrices generated by NumPy's `default_rng` with seeds 3901, 3902 and 3903. Each matrix has its trace removed and is normalized to unit Frobenius norm, then lifted to orbital space as the level matrix tensored with I_2 . Its exact two-quasiparticle field consists of the antisymmetric pair-creation block of the HFB-transformed one-body operator and its complex conjugate. The weak-paired modal comparison uses only the first dense field. A five-field weak-paired basis is constructed separately for the comparison between phase-specific and shared spaces.

Snapshots use 57 equally spaced energies $E = 0, 0.5, \dots, 28$ and Lorentzian full width $\Gamma = 0.25$, so the driven shift is $E + i\Gamma/2 = E + 0.125i$. Validation and test strengths use 700 equally spaced energies including 0 and 28. Specifically, the scored curve in this benchmark is the positive-frequency strength

$$S_a^+(E) = \sum_{\nu \text{ physical}, \omega_\nu > 0} \frac{\Gamma}{2\pi} \frac{|z_\nu^\dagger \mathcal{F}_a|^2}{(E - \omega_\nu)^2 + (\Gamma/2)^2}, \quad z_\nu^\dagger \mathcal{N} z_\nu = 1. \quad (36)$$

It omits the negative-frequency Lorentzian tail present in the full retarded response; full, reduced and PMM results are all scored with the same convention. The moments are sums $m_k = \sum_\nu \omega_\nu^k |z_\nu^\dagger \mathcal{F}_a|^2$ for $k = -1, 0, 1$, not quadratures over the finite plotting interval. The two eigenvalues of smallest absolute magnitude are excluded as the number-symmetry pair when the number-field norm exceeds 10^{-12} ; the moment sums additionally require $\omega_\nu > 10^{-8}$.

Complete driven-response vectors are divided by their Euclidean norms before collection. They are transported into the two-quasiparticle coordinates of the middle construction point using the unitary direct rotation between the HFB generalized-density projectors. The induced rotation acts on antisymmetric pair amplitudes and their conjugates, preserving the doubled metric and partner map; the minimum principal overlap required by this transport

TABLE I. HFB-QRPA parameter sets. Counts are construction/validation/test couplings. Test ranges give the smallest and largest saved test parameters, not a replacement for the full point arrays.

Case	f/g	Counts	Construction g	Test g
Weak normal	0.05	7/6/7	[0.20000, 0.68000]	[0.21370, 0.65790]
Weak paired	0.05	9/8/8	[0.82000, 2.00000]	[0.83710, 1.94710]
Strong paired	0.50	9/8/8	[0.42000, 2.00000]	[0.43730, 1.88390]

is 0.05. At evaluation the frozen basis is transported back to the local HFB coordinates before projection.

At every paired construction point the particle-number field $\mathcal{F}_N$ and its null direction $\mathcal{N}\mathcal{F}_N$ are normalized, transported, and supplied as required basis directions. Snapshot covariance eigenvalues below 10^{-12} times the largest are discarded. The required directions, retained POD directions and their partners are combined using a relative singular-value threshold 10^{-10} . The restricted-metric cutoff is 10^{-8} ; metric normalization uses 10^{-10} . Required directions precede the POD directions in the metric-orthogonalization sequence. Requested final ranks are

$$\{2, 4, 6, 8, 10, 12, 16, 20, 24, 28, 32, 40, 48, 56, 64, 72, \\ 80, 96, 112, 128, 144, 160, 176, 182\},$$

limited to the even dimensions supported by both signs of the restricted metric. Validation requires mean and maximum relative response errors at most 10^{-3} and 2×10^{-3} , respectively. It also requires zero stability failures and complex physical modes, Hermiticity and partner defects at most 10^{-10} and 10^{-9} , spectral-pairing defect at most 10^{-7} , number-null residual at most 10^{-8} , and number-pair displacement at most 10^{-5} . The smallest passing rank is retained. The three selected dimensions are 24, 56 and 56.

C. Compression and modal comparisons

Driven-response, unweighted-modal and field-weighted-modal bases are evaluated at the same final dimension in each case. The modal constructions use every positive-frequency

eigenvector returned at the same construction parameters, with no energy-window cutoff. Each eigenvector is first normalized to unit Euclidean norm. Unweighted modal POD gives these vectors equal weight. For field-weighted modal POD, a normalized mode v_ν is multiplied by

$$\frac{w_\nu}{\max_\mu w_\mu}, \quad w_\nu = \left(\sum_a |v_\nu^\dagger \mathcal{F}_a|^2 \right)^{1/2}, \quad (37)$$

where the sum runs over that case's construction fields and the maximum is taken separately at each construction parameter. The denominator is bounded below by 10^{-15} . Both modal collections undergo the same transport, number-direction augmentation, partner closure and metric normalization as the driven collection. The comparison uses the same physical test parameters and field family within each case. Table II gives the mean and maximum relative strength errors. All entries have zero reported stability failures. The field-weighted modal basis is retained as a separate comparison rather than grouped with unweighted modal POD.

The driven-response construction gives low maximum test errors in all three cases. Field-weighted modes give similarly accurate results in the paired cases, but a large error remains in the weak-normal case, while unweighted modes give large errors throughout. The driven-response construction therefore gives more consistent accuracy across these tests.

This is not a direct consequence of the POD optimality statement. That result concerns the average squared projection error of the sampled, normalized driven-response vectors, rather than the strength functions obtained from the reduced equations at withheld parameter values. Nevertheless, the selected spaces also reproduce these withheld strength functions accurately. The results therefore show that the sampled driven-response vectors contain information that remains useful beyond the points used to construct the reduced space.

For the driven-response bases, the largest test-set Hermiticity defect is 3.39×10^{-15} and the largest spectral-pairing defect is 1.49×10^{-9} . The particle-number null residual in the paired cases is at most 1.74×10^{-12} . No artificial complex physical modes are reported. Small numerical displacements of the particle-number pair are recorded separately from physical stability.

The interaction retains spin-rotation symmetry, including the term proportional to f , and all probes are spin scalar. For a spin-symmetric reference, the normal scalar particle-hole sector

TABLE II. Equal-rank HFB-QRPA test errors. The weak-normal and strong-paired cases use five fields; the weak-paired comparison uses the first dense field.

Case	Basis	r	Mean ε	Max. ε
Weak normal	Driven response	24	1.32×10^{-11}	1.38×10^{-10}
Weak normal	Unweighted modes	24	1.00×10^0	1.07×10^0
Weak normal	Field-weighted modes	24	2.80×10^{-1}	5.89×10^{-1}
Weak paired	Driven response	56	8.69×10^{-12}	1.37×10^{-11}
Weak paired	Unweighted modes	56	8.16×10^{-1}	9.23×10^{-1}
Weak paired	Field-weighted modes	56	1.43×10^{-13}	2.24×10^{-13}
Strong paired	Driven response	56	2.26×10^{-13}	1.13×10^{-12}
Strong paired	Unweighted modes	56	9.92×10^{-1}	1.35×10^0
Strong paired	Field-weighted modes	56	4.49×10^{-12}	1.83×10^{-11}

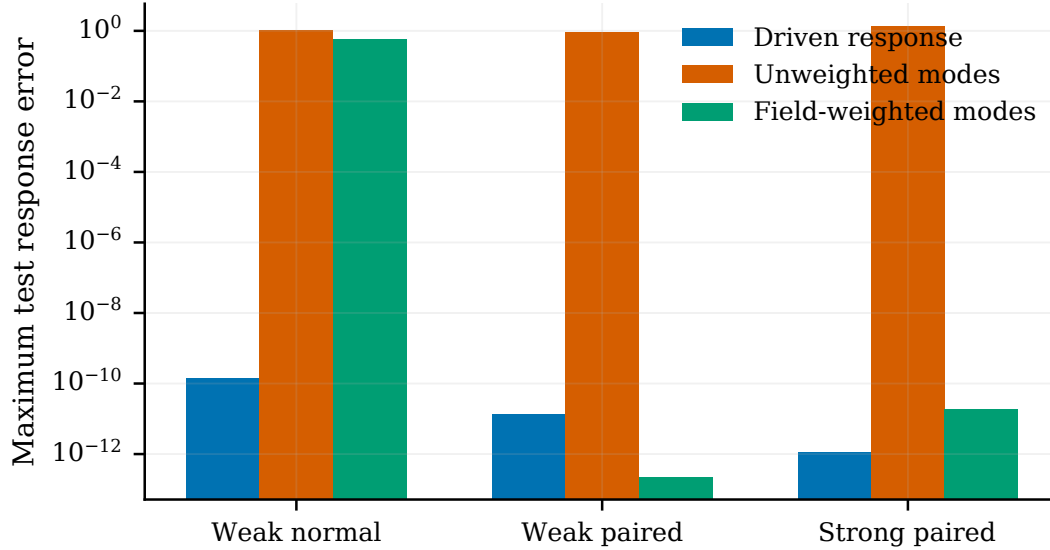


FIG. 1. Equal-rank modal comparison for the three HFB-QRPA cases. The plotted quantity is the maximum relative strength-function error on each case's test set. The ranks are 24, 56 and 56.

has doubled dimension $2(N_p/2)(L - N_p/2) = 24$. The paired spin-singlet two-quasiparticle sector has doubled dimension $L(L + 1) = 56$. The selected ranks coincide with these symmetry-allowed dimensions. Thus the nearly exact responses demonstrate recovery of the response-active sector; this benchmark does not establish further compression below

that sector's dimension. The sector was not supplied explicitly to the POD construction.

The shared-phase comparison uses the same five fields in both phases. Separate spaces require dimension 24 in the normal phase and 56 in the paired phase, with 80 basis columns stored in total. A shared space uses dimension 56 in both phases. The separate and shared constructions have maximum test errors 1.38×10^{-10} and 4.14×10^{-9} , respectively. Both are accurate; phase separation reduces the active normal-phase dimension rather than being necessary for accuracy in this comparison.

D. One- and two-parameter PMMs

The one-parameter PMM is trained on the strong-paired ray $f = 0.5g$, with $g \in [0.42, 2]$, using the frozen 56-dimensional five-field basis. The training pool consists of the 24 sorted Chebyshev roots mapped to this interval. More generally, the mapped nodes used here are

$$c_j^{(n)}(a, b) = \frac{a+b}{2} + \frac{b-a}{2} \cos \frac{(2j+1)\pi}{2n}, \quad j = 0, \dots, n-1. \quad (38)$$

Nested training subsets begin with the smallest and largest nodes and repeatedly add the node farthest from those already selected, resolving ties by the lower index in the sorted pool. The candidate training sizes are 8, 12, 16, 20 and 24, and degrees are 3, 4, $\dots$, 10, subject to $p+1 \leq n_{\text{train}}$. The eight strong-paired basis-validation couplings are also used for PMM validation; the eleven PMM test couplings are

$$\{0.4627, 0.5879, 0.7063, 0.8561, 1.0189, 1.1633, \\ 1.3097, 1.4729, 1.6481, 1.8127, 1.9733\}.$$

For each size, the degree minimizes the mean validation PMM error, followed by maximum error and degree to break ties. The smallest training size passing mean and maximum tolerances 10^{-3} and 2×10^{-3} is retained. The full-space PMM uses the same training operators and the degree chosen for the reduced PMM. Validation selects 16 training operators and polynomial degree 10. The corresponding mean and maximum validation errors are 4.16×10^{-4} and 9.97×10^{-4} .

The two-parameter PMM uses the physical strengths (g, f) as inputs. Sampling is specified on $g \in [0.85, 2]$ and $f/g \in [0.05, 0.50]$, with $f = (f/g)g$ when forming the Hamiltonian and

polynomial inputs. It reuses the strong-paired basis and all five fields without collecting new basis snapshots. Its separate basis check uses

$$\{0.85, 1.20, 1.60, 2.00\} \times \{0.05, 0.275, 0.50\}$$

in $(g, f/g)$ coordinates. The 64-point training pool is the Cartesian product of eight mapped Chebyshev nodes in each sampling coordinate, ordered with f/g outermost and g innermost. Nested subsets are selected by farthest-point sampling in the physical (g, f) coordinates after scaling each coordinate to $[0, 1]$: start with the point farthest from the scaled centre, then the point farthest from it, and subsequently maximize distance to the nearest selected point. Ties use the first index in the pool. PMM validation uses the twelve pairs

$$\{0.90, 1.11, 1.32, 1.53, 1.74, 1.95\} \times \{0.1625, 0.3875\}.$$

The fourteen test pairs, again written as $(g, f/g)$, are

$$\begin{aligned} &\{(0.8723, 0.0713), (0.9479, 0.1431), (1.0367, 0.2869), \\ &\quad (1.1183, 0.4571), (1.2079, 0.0947), (1.2961, 0.1943), \\ &\quad (1.3817, 0.3569), (1.4689, 0.4813), (1.5571, 0.1277), \\ &\quad (1.6423, 0.2399), (1.7317, 0.4031), (1.8169, 0.3187), \\ &\quad (1.9043, 0.1763), (1.9769, 0.4391)\}. \end{aligned}$$

Candidate training sizes are 20, 32, 48 and 64; total polynomial degrees are $1, 2, \dots, 6$ with $(p+1)(p+2)/2 \leq n_{\text{train}}$. Degrees are selected by the same validation ordering as above, and the smallest passing size is selected with mean and maximum tolerances 5×10^{-3} and 2×10^{-2} . Full and reduced PMMs again share the selected degree. Validation selects 48 operators and total polynomial degree 6, with mean and maximum validation errors 1.33×10^{-3} and 3.90×10^{-3} .

Table III compares the selected reduced and full-space PMMs. The mean compression errors of the reduced models are 1.85×10^{-13} and 1.98×10^{-13} for the one- and two-parameter tests. At the shown precision, PMM and total response errors are therefore the same.

Both selected reduced models have zero physical-stability failures and zero applied stability shift on the test sets. Their maximum particle-number null residuals are 6.42×10^{-17} and

TABLE III. Selected HFB-QRPA PMMs on the held-out data. The reduced dimension is 56 and the full dimension is 182. The one-parameter fit uses 16 matrices and degree 10; the two-parameter fit uses 48 matrices and total degree 6. The final column is the mean total relative m_1 error.

Parameters	Representation	Mean total	Max. total	Mean m_1
1 parameter	Reduced PMM	4.94×10^{-4}	1.68×10^{-3}	1.4×10^{-5}
1 parameter	Full PMM	4.89×10^{-4}	1.70×10^{-3}	1.4×10^{-5}
2 parameters	Reduced PMM	1.22×10^{-3}	3.16×10^{-3}	3.1×10^{-5}
2 parameters	Full PMM	1.14×10^{-3}	3.40×10^{-3}	2.8×10^{-5}

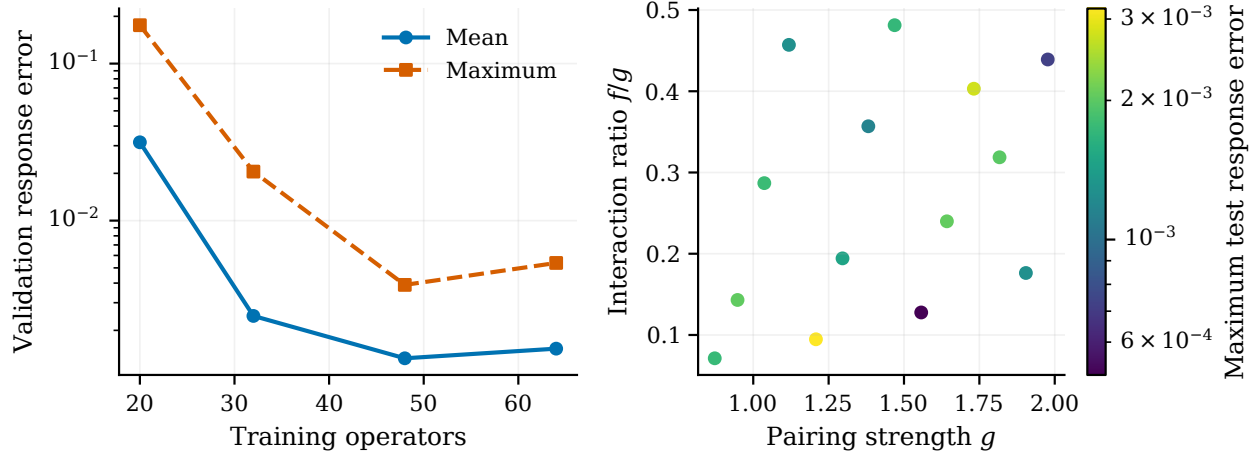


FIG. 2. Two-parameter HFB-QRPA PMM. Left: validation errors for the candidate training sizes. Right: maximum error over the tested fields at each held-out parameter pair for the selected reduced PMM. The plot uses g and f/g as sampling coordinates; the model inputs are the physical strengths (g, f) .

6.05×10^{-17} . Their maximum physical imaginary parts are 3.15×10^{-14} and 4.61×10^{-14} . The reduced particle-number frequencies remain within 4.02×10^{-8} of zero across these tests.

The two-parameter fit stores 87 808 reduced coefficient-array entries, compared with 927 472 full-space entries, a factor of 10.5625. Counting independent real coefficients instead gives a factor of 10.4342. These are different counts and are not used interchangeably.

TABLE IV. Half-filled matrix-action HF-RPA systems. D includes both forward and backward particle-hole amplitudes. The scalar-sector dimension is a separate reference for compression inside the active channel.

L/N_p	D	Scalar sector	d
12/12	288	72	0.545455
16/16	512	128	0.400000
20/20	800	200	0.315789

V. RESTRICTED-INFORMATION MATRIX-ACTION HF-RPA

In this section we construct the reduced space without complete driven RPA responses. The physical model is Eq. (34), but the reference is obtained with ordinary HF and the response is RPA. The available information consists of the HF reference, independent particle-hole response solves, actions of the residual stability matrix and the physical external field. The full RPA matrix, its eigensystem and complete driven solutions are not used to construct or select the basis.

A. Systems and information access

We use the three half-filled systems in Table IV. The one-particle bandwidth is fixed at 6, so $d = 6/(L - 1)$. The varied coordinate is $g/d \in [0.2, 2.0]$, and $f = 0.5g$. The full doubled particle-hole dimension is $D = 2N_p(2L - N_p) = 2L^2$ at half filling.

The separate parameter sets contain 13 basis-construction points, 12 rank-validation points, 16 available PMM-training points, 16 PMM-validation points and 20 test points. In the dimensionless coordinate g/d , the first three arrays are

$$\begin{aligned}
0.200 + 0.150j, & \quad j = 0, \dots, 12, \\
0.275 + 0.150j, & \quad j = 0, \dots, 11, \\
0.21875 + 0.1175j, & \quad j = 0, \dots, 15,
\end{aligned} \tag{39}$$

respectively. The PMM-validation points are $1.1 + 0.9\xi_j$, where ξ_j are the 16 roots of the Legendre polynomial of degree 16. The test points are $1.1 - 0.9\cos[(2j - 1)\pi/40]$,

$j = 1, \dots, 20$. Both latter arrays are rounded to ten decimal places. All five arrays are disjoint.

The construction energies are $E_j = 8j/7$, $j = 0, \dots, 7$ (stored to twelve decimal places). Rank validation uses $E_j = 0.125 + 0.25j$, $j = 0, \dots, 31$, and response evaluation uses 500 equally spaced energies including 0 and 8. The Lorentzian full width is $\Gamma = 0.25$, hence $\zeta(E) = E + 0.125i$. These energies and widths are in the same units as the fixed one-particle bandwidth. The probe is a spin-scalar nearest-neighbour field,

$$(F^{(1)})_{p\sigma, q\sigma'} = \frac{\delta_{\sigma\sigma'} \delta_{|p-q|,1}}{\sqrt{2(L-1)}}. \quad (40)$$

Its level-space matrix has unit Frobenius norm. In local HF orbitals C_o and C_v , spanning the occupied and unoccupied spaces, the doubled field is

$$\mathcal{F} = \begin{pmatrix} \text{vec}(C_v^\dagger F^{(1)} C_o) \\ \text{vec}(C_v^\dagger F^{(1)} C_o)^* \end{pmatrix}, \quad (41)$$

with the hole index running fastest.

The scalar-sector dimensions in Table IV are reported separately from the full dimensions. This distinction separates removal of inactive spin channels from compression inside the active response sector. The sector is checked only in the subsequent full-order evaluation, at the first test coupling of each system. Across these three checks its field leakage is below 6.76×10^{-15} and its invariance defect is below 1.67×10^{-15} .

B. Action-generated directions and rank selection

Let $\mathcal{H}_0$ denote the independent particle-hole stability operator obtained from the HF reference, and let $\mathcal{V} = \mathcal{H} - \mathcal{H}_0$ denote the residual interaction. At each construction energy, the first vector is the independent response

$$[\mathcal{H}_0 - \zeta(E)\mathcal{N}]s_0 = -\mathcal{F}. \quad (42)$$

Here $\mathcal{H}_0 = \text{diag}(\Delta, \Delta)$, where Δ is diagonal in the particle-hole index (m, i) , with entries $\epsilon_m - \epsilon_i$ from the HF orbital energies. Its shifted inverse is evaluated elementwise, using denominators $\epsilon_m - \epsilon_i - \zeta$ and $\epsilon_m - \epsilon_i + \zeta$ for the two amplitude blocks. The two correction

directions are

$$[\mathcal{H}_0 - \zeta(E)\mathcal{N}]s_{j+1} = -\mathcal{V}s_j, \quad j = 0, 1. \quad (43)$$

The three increments s_0, s_1, s_2 are retained separately; no cumulative approximate response is used as a snapshot. At each parameter–energy pair, all three are divided by the same factor $\max(\|s_0\|_2, 10^{-15})$. Thus the higher-order directions retain their sizes relative to the independent response, rather than each receiving unit weight. The residual action contracts the transition density $C_v X C_o^\dagger + C_o Y^\top C_v^\dagger$ with the factorized two-body interaction and transforms the induced field back to particle–hole coordinates. This is an analytic matrix action, not a finite-difference calculation or a full RPA solve.

The common HF anchor is $g_* = 1.10d$. With occupied projectors $P_o(g) = C_o(g)C_o(g)^\dagger$, let

$$R_g = \text{polar}[P_o(g_*)P_o(g) + (I - P_o(g_*))(I - P_o(g))], \quad (44)$$

where polar denotes the unitary polar factor. The occupied and unoccupied coordinate rotations are

$$T_o = C_o(g_*)^\dagger R_g C_o(g), \quad T_v = C_v(g_*)^\dagger R_g C_v(g).$$

Transport to the anchor acts as

$$(X, Y) \mapsto (T_v X T_o^\dagger, T_v^* Y T_o^\top). \quad (45)$$

The smallest singular values of both occupied and unoccupied inter-reference overlaps must be at least 0.10. The transported, scaled increments give $13 \times 8 \times 3 = 312$ snapshot columns. POD orders their directions, after which partner closure and metric normalization produce the candidate spaces. The same unitary transport is inverted when a basis is used at a new coupling.

Rank selection uses a residual-based output indicator. For a lifted reduced response x_r , the available residual and its independent-particle correction are

$$\mathfrak{r} = -\mathcal{F} - [\mathcal{H} - \zeta(E)\mathcal{N}]x_r, \quad [\mathcal{H}_0 - \zeta(E)\mathcal{N}]\delta x_0 = \mathfrak{r}. \quad (46)$$

At each of the 12 rank-validation couplings, define the real curves

$$\begin{aligned} S_r(E) &= -\pi^{-1} \text{Im}(\mathcal{F}^\dagger x_r(E)), \\ \delta S_0(E) &= -\pi^{-1} \text{Im}(\mathcal{F}^\dagger \delta x_0(E)), \end{aligned} \quad (47)$$

and the indicator

$$I(g) = \frac{\|\delta S_0\|_2}{\max(\|S_r\|_2, 10^{-15})}. \quad (48)$$

Both norms are unweighted Euclidean norms over the 32 rank-validation energies. The mean and maximum are then taken over couplings, not individual energies. The curve in Eq. (47) is the shifted-equation contraction, rather than the positive-mode Lorentzian sum used in the reported test errors. The tighter criterion uses mean and maximum limits 10^{-3} and 2×10^{-3} , respectively; the relaxed criterion uses 5×10^{-3} and 10^{-2} . The smallest generated candidate satisfying both limits is selected, provided every validation matrix obeys $\lambda_{\min}(\mathcal{H}_r) \geq -10^{-8}$. The indicator estimates the output correction under the independent-particle inverse; it is not a rigorous bound on the unavailable complete response. All RPA matrices diagonalized in this rank test are reduced matrices.

C. Field augmentation and the sum rule

The third construction uses the relaxed indicator and retains the positive-signature part $(\mathcal{F} + \mathcal{N}\mathcal{F})/2$ of the physical field at the existing 13 construction couplings. Each nonzero direction is transported to the anchor and normalized in the Euclidean norm. Its partner is added, so both the field and its metric transform are retained at the construction points. These required directions precede the POD directions in metric normalization; their partner-closed span has dimension 18 in the stored runs. No complete response or additional residual-action data are required to generate them. If Q_r is a Euclidean-orthonormal basis for the final anchor space and $w(g)$ is the transported $\mathcal{N}\mathcal{F}(g)$, the additional acceptance condition is

$$\max_{g \text{ in rank validation}} \frac{\|(I - Q_r Q_r^\dagger)w(g)\|_2}{\max(\|w(g)\|_2, 10^{-15})} \leq 10^{-8}. \quad (49)$$

The relevant preservation condition is $\mathcal{N}\mathcal{F} \in S_{\text{ret}}$. It preserves the quadratic expression entering the energy-weighted sum rule. Indeed, if $w = \mathcal{N}\mathcal{F} = U_r c$, then

$$w^\dagger \mathcal{H} w = c^\dagger \mathcal{H}_r c, \quad c = J_r \mathcal{F}_r. \quad (50)$$

With the present field convention, $m_1 = \frac{1}{2} w^\dagger \mathcal{H} w$. Thus exact projection does not change this expression when the inclusion holds. Representation between the construction couplings is checked by Eq. (49) and by the subsequent test moments, rather than assumed exact. A

TABLE V. Post-freeze matrix-action RPA tests. Each row contains 20 test couplings. “Sum-rule” denotes the relaxed indicator with required physical field directions. The last column concerns exact compression, without PMM interpolation.

L/N_p	Construction	r	Mean ε	Max. ε	Mean m_1 error
12/12	Tighter	56	7.71×10^{-4}	8.47×10^{-4}	3.70×10^{-4}
12/12	Relaxed	40	3.27×10^{-3}	5.30×10^{-3}	8.60×10^{-2}
12/12	Sum-rule	48	2.44×10^{-3}	3.25×10^{-3}	6.32×10^{-13}
16/16	Tighter	80	9.78×10^{-4}	1.18×10^{-3}	8.00×10^{-5}
16/16	Relaxed	56	3.73×10^{-3}	5.00×10^{-3}	7.42×10^{-4}
16/16	Sum-rule	56	4.18×10^{-3}	5.43×10^{-3}	1.07×10^{-12}
20/20	Tighter	144	5.11×10^{-4}	7.34×10^{-4}	9.10×10^{-5}
20/20	Relaxed	72	2.95×10^{-3}	3.49×10^{-3}	4.57×10^{-3}
20/20	Sum-rule	64	3.98×10^{-3}	5.26×10^{-3}	2.68×10^{-13}

PMM replaces $\mathcal{H}_r$ by an approximation and can still introduce moment error. Compression and PMM moment errors are therefore recorded separately.

D. Frozen full-RPA tests

The basis, dimension, polynomial degrees and PMM coefficients are fixed before the full RPA calculation is enabled. The full-RPA assembly/eigensolve interface is disabled during construction and model selection, and its build counter is asserted to be zero immediately before enabling test evaluation. HF orbital solves, independent particle-hole solves, matrix actions and reduced solves remain available throughout. The three basis variants are then evaluated on the same 20 test couplings. Table V reports the relative strength and energy-weighted moment errors. No stability failures occur in these tests.

The sum-rule-preserving construction selects ranks 48, 56 and 64. For the largest system this is $800 \rightarrow 64$ relative to the full doubled space and $200 \rightarrow 64$ within the scalar sector. The largest test-set spectral-pairing defect for the sum-rule-preserving spaces is 3.45×10^{-14} .

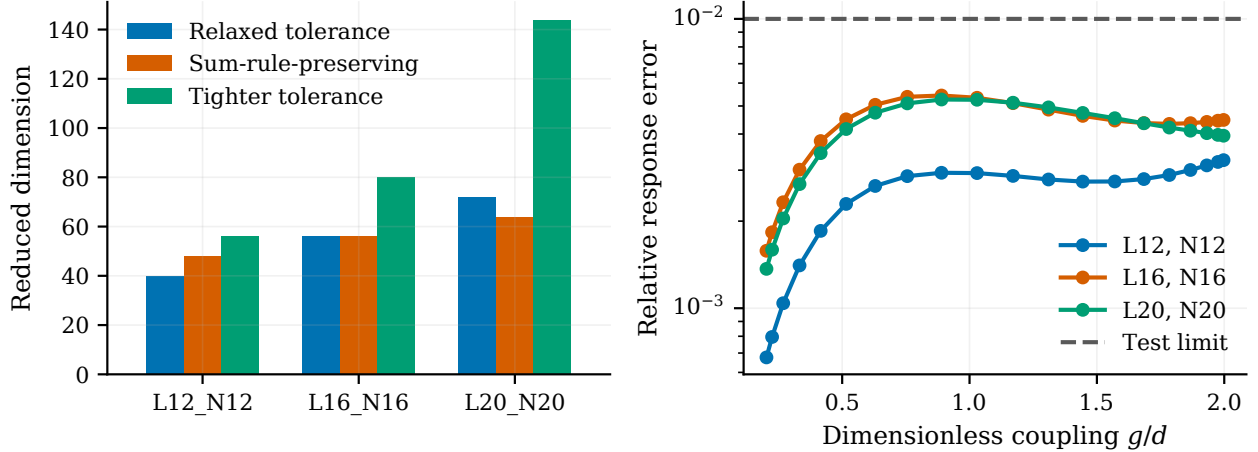


FIG. 3. Restricted-information HF-RPA reduction. Left: final dimensions for the three rank-selection rules. Right: test response errors of the field-augmented, sum-rule-preserving construction. Complete RPA responses are used only after construction and model selection have been frozen. The horizontal line is the declared post-freeze test limit, not a full-response rank-selection criterion.

E. PMM learning in the action-generated space

For $L = 20$ and $N_p = 20$, projected training matrices are formed by applying the full stability operator to the retained basis vectors. This requires matrix actions, not full matrix assembly. The physical field remains projected from the HF reference. The PMM-training sizes are 4, 6, 8, 12 and 16, using nested farthest-point subsets of the 16-point training pool. Degrees $p \in \{1, 2, 3, 4\}$ with $p+1 \leq n_{\text{train}}$ are compared on the 16 PMM-validation couplings and the 500-point response grid. The reference for this comparison is the action-projected reduced model. Degree selection minimizes, in order: whether any validation point requires a stability shift, mean relative response error, maximum relative response error, and degree. The selected degree is 3 for four training operators and 4 for the other sizes. Table VI shows the learning curve for the rank-64 sum-rule-preserving construction; it does not select a new training size from the test results.

All reported action PMMs have zero stability failures and zero applied stability corrections. With degree 4, the rank-64 generic complex structured model has 10 400 independent real coefficients. Relative to hypothetical models of the same degree in that same matrix class, the reductions are 154.04 for the full 800-dimensional representation and 9.66 for the 200-dimensional scalar representation. These are representation counts, not comparisons with

TABLE VI. Rank-64 PMM learning curve for $L = N_p = 20$. The mean compression error is 3.981×10^{-3} for every row. The last column is the relative m_1 error of the PMM response against the full RPA response.

Matrices	p	Mean PMM	Mean total	Max. total	Mean m_1 error
4	3	2.15×10^{-3}	4.39×10^{-3}	6.20×10^{-3}	2.28×10^{-3}
6	4	6.78×10^{-4}	4.06×10^{-3}	5.42×10^{-3}	5.99×10^{-4}
8	4	6.74×10^{-4}	4.00×10^{-3}	5.67×10^{-3}	6.18×10^{-4}
12	4	6.08×10^{-4}	4.08×10^{-3}	5.54×10^{-3}	5.43×10^{-4}
16	4	6.25×10^{-4}	4.07×10^{-3}	5.66×10^{-3}	5.65×10^{-4}

trained full-space PMMs in this experiment; an additional reality constraint would reduce the absolute counts.

VI. MATRIX-FREE DIRQFAM APPLICATION TO ^{20}Ne

This section applies the same compression–PMM construction when the full stability matrix is unavailable. Complete response amplitudes are obtained from a small number of FAM calculations. Stability-matrix actions then provide the projected training matrices without constructing the full matrix or its eigensystem.

A. Backend and physical setting

FAM evaluates the induced mean-field response needed for the RPA/QRPA equations without assembling all residual-interaction matrix elements [7, 8]. The relativistic implementation uses a self-consistent relativistic Hartree–Bogoliubov reference [9]. We use DIRQFAM v2.0.0 with additional response-amplitude and stability-action output [10, 11].

The application is the isoscalar quadrupole response of deformed ^{20}Ne with the DD-ME2 functional. The retained response model has angular-momentum projection $K = 0$. Ordinary FAM curves for $K = 0, 1, 2$ are evaluated at the common reference as a channel check, but no reduced $K = 1$ or $K = 2$ model is used in the reported deformation study. The backend supports pairing, but the reference in this ^{20}Ne calculation has zero pairing

energy. The physical calculation is therefore in the relativistic RPA limit. The oscillator basis contains six large-component and seven small-component fermion shells and 50 meson shells. Its deformation and the initial mean-field deformation remain fixed at 0.547 while a quadrupole constraint varies the self-consistent deformation; the input constraint stiffness is 0.010. The parameter β denotes the target total quadrupole deformation, with the reported converged total required to agree within 5×10^{-7} . It is not the deformation of the oscillator basis. The quadrature uses 25 Gauss–Hermite nodes on the positive axial half-line and 50 Gauss–Laguerre radial nodes. Coulomb and finite-range separable pairing are enabled, with attractive pairing strength -728 MeV fm^3 for each species, range 0.644204936 fm, and initial neutron and proton gaps of 1 MeV. Pairing vanishes in the converged reference rather than being switched off. The response solver uses GMRES with relative residual tolerance 10^{-9} and at most 70 Arnoldi vectors; completed runs must also satisfy ground-state and response residual limits of 10^{-6} and 10^{-8} , respectively.

As an independent backend check, the adapted code is tested against the official paired, deformed ^{152}Sm example from the v2.0.0 distribution. The release and study executables give identical stored strengths, with and without the interface. The release differs from the distributed strength output by 6.09×10^{-10} in relative L^2 norm. Amplitude export and matrix-action checks also pass. Appendix A gives the settings and diagnostics. This paired backend check is separate from the unpaired ^{20}Ne reduction experiment.

B. Response sampling and common coordinates

Each deformation β has a different self-consistent quasiparticle coordinate frame. We use the anchor deformation $\beta_0 = 0.547$. Let $T_{\beta \leftarrow \beta_0}$ denote the implemented coordinate transport from the anchor frame to the local frame. Response amplitudes are transported to the anchor before normalization and POD. The basis is transported back for local projection,

$$U_r(\beta) = T_{\beta \leftarrow \beta_0} U_r. \quad (51)$$

This makes it possible to combine response vectors from different deformations in one snapshot matrix. It does not require identifying their untransformed coordinates as the same vector. For each nucleon species, let (u_0, v_0) and (u_β, v_β) be the full-precision Bogoliubov

blocks in the common one-body basis. The implementation forms

$$\begin{aligned} O_\beta &= u_0^\dagger u_\beta + v_0^\dagger v_\beta, & q_\beta &= \text{polar}(O_\beta)^\dagger, \\ X_\beta &= q_\beta X_0 q_\beta^\dagger, & Y_\beta &= q_\beta^* Y_0 q_\beta^\top. \end{aligned} \tag{52}$$

Here X and Y are the matrix-shaped forward and backward amplitude blocks before selecting the allowed two-quasiparticle coordinates; $\text{polar}(O) = LR^\dagger$ when $O = L\Sigma R^\dagger$. These block operations represent the direct rotation of the generalized-density projectors without constructing their full tangent matrices. The reverse transport uses $q_\beta^\dagger$.

The smallest singular value of O_β must exceed 10^{-6} . Relative Bogoliubov normalization and polar-gauge unitarity defects are limited to 10^{-10} and 10^{-9} , respectively. Selecting the allowed pair coordinates must give omitted-coordinate, metric and partner defects below 10^{-8} . Round-trip and probe metric-Gram defects are also recorded. Thus preservation by the packed transport is checked on the represented amplitudes, not inferred from the unitarity of the polar factor alone.

Five construction deformations are used,

$$\{0.500, 0.525, 0.547, 0.575, 0.600\}, \tag{53}$$

and the five rank-validation deformations are

$$\{0.510, 0.530, 0.550, 0.570, 0.590\}. \tag{54}$$

The 20 energies $E_j = 2.5j$ MeV, $j = 1, \dots, 20$, supply the complete complex FAM response at each construction deformation. The evaluation grid runs from 0 to 50 MeV in steps of 0.5 MeV. Both the backend and reduced solver use $z = E + i0.25$ MeV: 0.25 MeV is the imaginary shift, corresponding to a Lorentzian full width of 0.50 MeV. In the exported convention the driven equation has right-hand side $-\mathcal{F}$, and the strength is $-\text{Im}(\mathcal{F}^\dagger x)/\pi$.

The final space is partner closed and metric canonicalized. The smallest accepted rank must have mean validation relative L^2 error at most 5×10^{-3} and maximum peak-scaled absolute error at most 10^{-2} . The latter is $\max_E |S_r(E) - S(E)| / \max_E |S(E)|$, maximized over validation deformations; it is not a pointwise relative error near strength-function zeros. Validation selects dimension $r = 100$ from the full doubled dimension $D = 35\,088$. The

retained fraction is $100/35088 \simeq 0.00285$. The mean validation response error is 3.632×10^{-3} . The canonical metric defect is 2.63×10^{-14} , the reported partner-closure defect is zero, and no complex modes occur in the validation calculations.

C. Matrix-free projection and PMM fitting

The added interface returns the frequency-independent stability action $v \mapsto \mathcal{H}(\beta)v$. The projected matrices and field are

$$\begin{aligned}\mathcal{H}_r(\beta) &= U_r(\beta)^\dagger \mathcal{H}(\beta) U_r(\beta), \\ \mathcal{N}_r(\beta) &= U_r(\beta)^\dagger \mathcal{N} U_r(\beta), \quad \mathcal{F}_r(\beta) = U_r(\beta)^\dagger \mathcal{F}(\beta).\end{aligned}\tag{55}$$

Only the r columns of $U_r(\beta)$ are passed through the stability-action interface. In particular,

$$\mathcal{H}_r(\beta) = U_r(\beta)^\dagger [\mathcal{H}(\beta)u_1(\beta) \cdots \mathcal{H}(\beta)u_r(\beta)].\tag{56}$$

The dense $35\,088 \times 35\,088$ matrix is never assembled. The actions are processed in bounded blocks. Each external action batch also prepares the reference and one single-frequency response before applying its supplied columns; the measured projection cost includes this setup. A raw projected matrix must have relative Hermiticity defect at most 10^{-7} , after which it is replaced by its Hermitian part. Subsequent Hermiticity diagnostics therefore describe the symmetrized matrix. The reduced metric is checked across all 16 available training points against the metric at the lowest deformation, using relative Frobenius norm. The maximum drift is 2.08×10^{-15} , below the 10^{-8} threshold for a constant representation. For each fitting subset, the fixed metric is the Hermitian, partner-odd projection of the arithmetic mean of its projected metrics; it is not reset to J_r . Its balanced, nonsingular signature is checked at relative tolerance 10^{-9} .

Candidate PMM training sizes are 4, 6, 8, 12 and 16. They are prefixes of the farthest-point ordering

$$\begin{aligned}(0.502, 0.598, 0.542, 0.568, 0.522, 0.582, 0.512, 0.532, \\ 0.558, 0.588, 0.506, 0.518, 0.538, 0.562, 0.578, 0.594).\end{aligned}$$

The endpoints are retained first; each subsequent point maximizes its distance to the nearest retained point, with ties choosing the smaller deformation. Chebyshev scaling uses these

TABLE VII. Held-out ^{20}Ne responses with $r = 100$. Every error in this table is a relative discrete L^2 error, with the reference for each stage specified in Eq. (31).

Deformation β	$\varepsilon_{\text{comp}}$	ε_{PMM}	$\varepsilon_{\text{total}}$
0.503636	2.626×10^{-3}	3.900×10^{-5}	2.623×10^{-3}
0.516059	4.274×10^{-3}	6.150×10^{-4}	4.267×10^{-3}
0.537577	3.170×10^{-3}	8.900×10^{-5}	3.171×10^{-3}
0.562423	3.114×10^{-3}	4.700×10^{-5}	3.114×10^{-3}
0.583941	4.192×10^{-3}	1.670×10^{-4}	4.197×10^{-3}
0.596364	3.628×10^{-3}	8.700×10^{-5}	3.628×10^{-3}
Mean	3.501×10^{-3}	1.74×10^{-4}	3.500×10^{-3}

fitting endpoints, giving centre 0.550 and half-span 0.048. At each size, admissible degrees from 1–4 minimize the mean relative L^2 error against exact reduced responses at the same five validation deformations as rank selection; exact ties choose the lower degree. The smallest training size with mean error at most 5×10^{-3} and maximum peak-scaled absolute error at most 2×10^{-2} is retained. The selected model uses four matrices, at deformations

$$\{0.502, 0.598, 0.542, 0.568\}, \quad (57)$$

and polynomial degree 3. It contains 20 200 independent real coefficients. The physical external field is projected at evaluation and is not learned.

D. Held-out responses

The six test deformations are disjoint from construction, rank validation and matrix fitting. Table VII reports the three relative L^2 errors. The mean total error is 3.50×10^{-3} and the largest test-point relative L^2 error is 4.267×10^{-3} . The mean PMM error relative to exact reduction is 1.74×10^{-4} .

For FAM, the moments are evaluated by the composite trapezoidal rule applied to $E^k S(E)$, $k = -1, 0, 1$, using the same grid for each method. The routine excludes energies $E \leq 10^{-8}$ MeV for all three moments, so the effective integration interval here is 0.5–50 MeV; no value of $S(E)/E$ is assigned at zero. The maximum reported total moment errors across the six

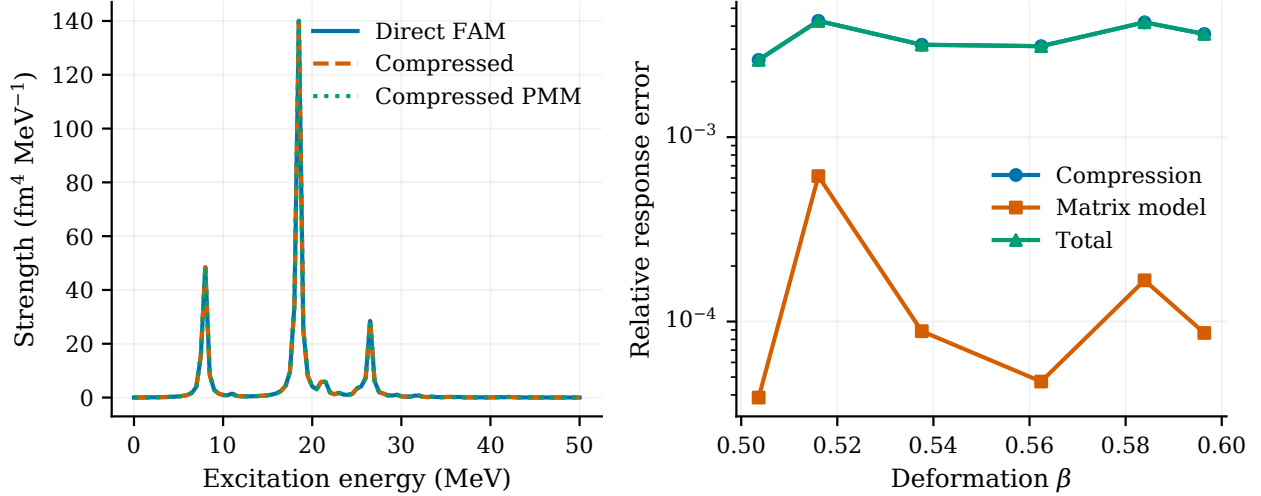


FIG. 4. The $K = 0$ isoscalar quadrupole response of ^{20}Ne . Left: direct FAM, exact compression and the reduced PMM at the held-out deformation $\beta = 0.562423$. Right: compression, PMM and total relative L^2 errors at all six test deformations. The response curves use the 0–50 MeV evaluation grid.

test deformations are 3.90×10^{-4} for m_{-1} , 6.63×10^{-4} for m_0 and 1.012×10^{-3} for m_1 . These are finite-grid response moments, not complete-spectrum sum rules. In every stage, the maximum scaled response error uses the peak absolute strength of its reference curve as denominator: direct FAM for compression and total error, and exact compression for PMM error.

Both exact and learned reductions have zero failed test points and zero applied stability shift. The learned matrices have spectral-pairing defect at most 1.03×10^{-14} and imaginary parts at most 3.11×10^{-14} . The minimum recorded stability eigenvalue of the learned reduced matrices is 6.451134; it is not a full-space excitation energy. These checks concern the specified field, nucleus and deformation interval.

E. Online cost and break-even

The complete PMM workload obtains a new self-consistent reference and exact physical field, loads that field, constructs the coordinate transport, projects the field, evaluates the small matrix model and solves the reduced response. The present export route obtains the reference and field through a one-frequency FAM calculation at 2.5 MeV; this full-space

TABLE VIII. Recorded online workload times. The conditional last row excludes preparation and loading of the physical reference/field and is not used for the complete speedup.

Workload	Time (s)	Complete speedup
Direct FAM, including output loading	103.1512	1.00
Reduced PMM with new exact field	3.8879	26.53
Reduced calculation after field preparation	0.4367	—

shifted solve is included in the online cost. The comparison is with a direct FAM response curve. Conditional timings after field preparation are recorded separately.

Timing is performed at $\beta = 0.562423$ on the 81-point grid from 0 to 40 MeV in steps of 0.5 MeV, with imaginary shift 0.25 MeV. One warm-up is excluded before five measured repetitions. The timed executable links OpenBLAS, and the external cases run in isolated processes with one numerical thread; the Python reduced workloads use the same thread limit and repetition policy. Completed external cases retain their original measured solver run-times when reused, rather than timing a cache lookup. Output loading is timed separately. The reported complete costs are sums of the median solver, loading and reduced-workload times, not medians of five jointly timed end-to-end runs. The environment is Apple arm64, 12 logical CPUs, Python 3.13.5, NumPy 2.1.3 and SciPy 1.15.3. Table VIII reports the resulting workload times. The complete online speedup is 26.53. The conditional reduced time is not used for this headline comparison.

Let T_{off} be the offline cost, T_{full} the direct FAM time and T_{PMM} the complete reduced-PMM time. For $T_{\text{full}} > T_{\text{PMM}}$, the break-even count is

$$N_{\text{BE}} = \left\lceil \frac{T_{\text{off}}}{T_{\text{full}} - T_{\text{PMM}}} \right\rceil. \quad (58)$$

Constructing the selected basis and matrix model costs 304.49 s, corresponding to four subsequent curves. This assumes that the rank, training size, polynomial degree and metric treatment have already been chosen. Including the basis and PMM selection calculations gives 1672.86 s and 17 subsequent curves. If the basis has already been supplied, the matrix-model construction costs 111.86 s and breaks even after two curves. These are different offline accounting choices, not different accuracy results. Backend validation, test-set evaluation

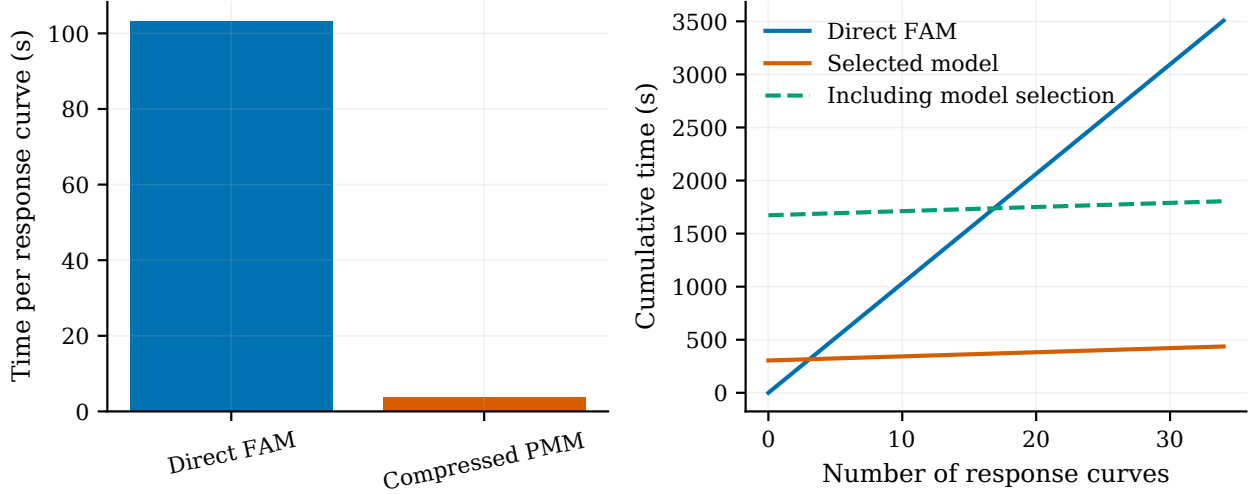


FIG. 5. Recorded FAM workload comparison. Left: complete time per direct and reduced-PMM response curve, including preparation of the exact physical field for the latter. Right: accumulated time with selected-model construction and with model selection included. The corresponding break-even counts are 4 and 17 curves.

and the timing campaign itself are excluded from all three offline totals.

VII. DISCUSSION, CONCLUSION AND OUTLOOK

Appendix A: DIRQFAM backend validation

The backend is checked independently of compression and PMM fitting. Three executables are compared: the unmodified DIRQFAM v2.0.0 release, the study revision without the added interface and the same revision with the interface. The release revision is 3e796299205e; the study revision is fb5b290020d5. They use GNU Fortran 15.3.0 with flags `-cpp`, `-ffree-line-length-none` and `-O3`. Linking uses the macOS SDK through `-isysroot` and Apple Accelerate through `-framework Accelerate`. Numerical-library thread counts are fixed to one and dynamic OpenMP adjustment is disabled. Source and executable hashes are recorded in the validation notebook and build manifests.

The official `test2/subtest1` example is the prolate paired ^{152}Sm calculation distributed with v2.0.0 [11, 12]. Its supplied input is retained unchanged. It uses DD-ME2, the isoscalar $J = 2$, $K = 2$ channel, finite-range separable pairing with attractive strength -728 MeV fm^3

for each species and range 0.644204936 fm, 14 large-component and 15 small-component fermion shells, and 50 meson shells. Coulomb and pairing terms are included; both initial pairing gaps are 1 MeV. The quadrature uses 25 Hermite and 50 Laguerre nodes. The oscillator and initial mean-field deformations are 0.288 with no quadrupole constraint. The converged deformation is 0.287721 and the total pairing energy is -4.897644 MeV. The response grid is 0–50 MeV in 2.5 MeV steps, with imaginary shift 0.25 MeV, GMRES relative tolerance 10^{-6} and 70 Arnoldi vectors. These are the distributed paired-reference settings, not the six-shell, constrained $K = 0$ deformation scan of Sec. VI.

The distributed reference strength and the release calculation differ by 6.09×10^{-10} in relative L^2 norm. The release and both study executables give the same stored strength, deformation and pairing energies. Exported amplitudes at 10, 15 and 40 MeV reconstruct the ordinary response. The official smaller ^{20}Ne `test3` example supplies the multi-vector stability-action tests; this is separate from the paired ^{152}Sm ordinary-response comparison. Those tests use a response at 25 MeV, two complex Gaussian probes with seed 20260820, and linear-combination, partner and field probes. They are sampled algebraic checks at one reference, not uniform operator-norm bounds.

All 24 declared checks pass. Reported zero differences refer to the stored outputs and the stated file comparisons. Amplitude export leaves the ordinary response unchanged, matrix-action calls leave the response files unchanged, and isolated-process repetitions agree. The latter checks use separate processes and do not test state reuse within a persistent solver process. The cached production timing cases and the validation rebuild record the same study source revision but different executable hashes; this comparison therefore does not constitute a byte-identical rerun of the timed executable.

TABLE IX. Selected DIRQFAM validation diagnostics and their declared limits. All rows pass. The omitted eight rows compare deformation and pairing energies between builds; all eight differences are zero in the saved output. The last three checks concern unchanged ordinary outputs or files.

Check	Value	Limit
Distributed/release strength	6.09×10^{-10}	1.0×10^{-3}
Release/study with interface	0	1.0×10^{-3}
Release/study without interface	0	1.0×10^{-3}
Study without/with interface	0	1.0×10^{-10}
Maximum GMRES residual	9.79×10^{-7}	1.1×10^{-6}
Amplitude/ordinary strength	2.51×10^{-9}	1.0×10^{-4}
Amplitude field contraction	3.56×10^{-13}	1.0×10^{-10}
Amplitude strength reconstruction	3.72×10^{-13}	1.0×10^{-12}
^{20}Ne distributed/interface strength	0	1.0×10^{-5}
Shifted-equation residual	3.42×10^{-8}	1.0×10^{-7}
Complex linearity	1.46×10^{-16}	1.0×10^{-12}
Hermitian action	3.56×10^{-10}	1.0×10^{-8}
Partner action	5.40×10^{-18}	1.0×10^{-12}
Ordinary response after export	0	0
Response files after action calls	0	0
Isolated-process repeat	0	1.0×10^{-12}

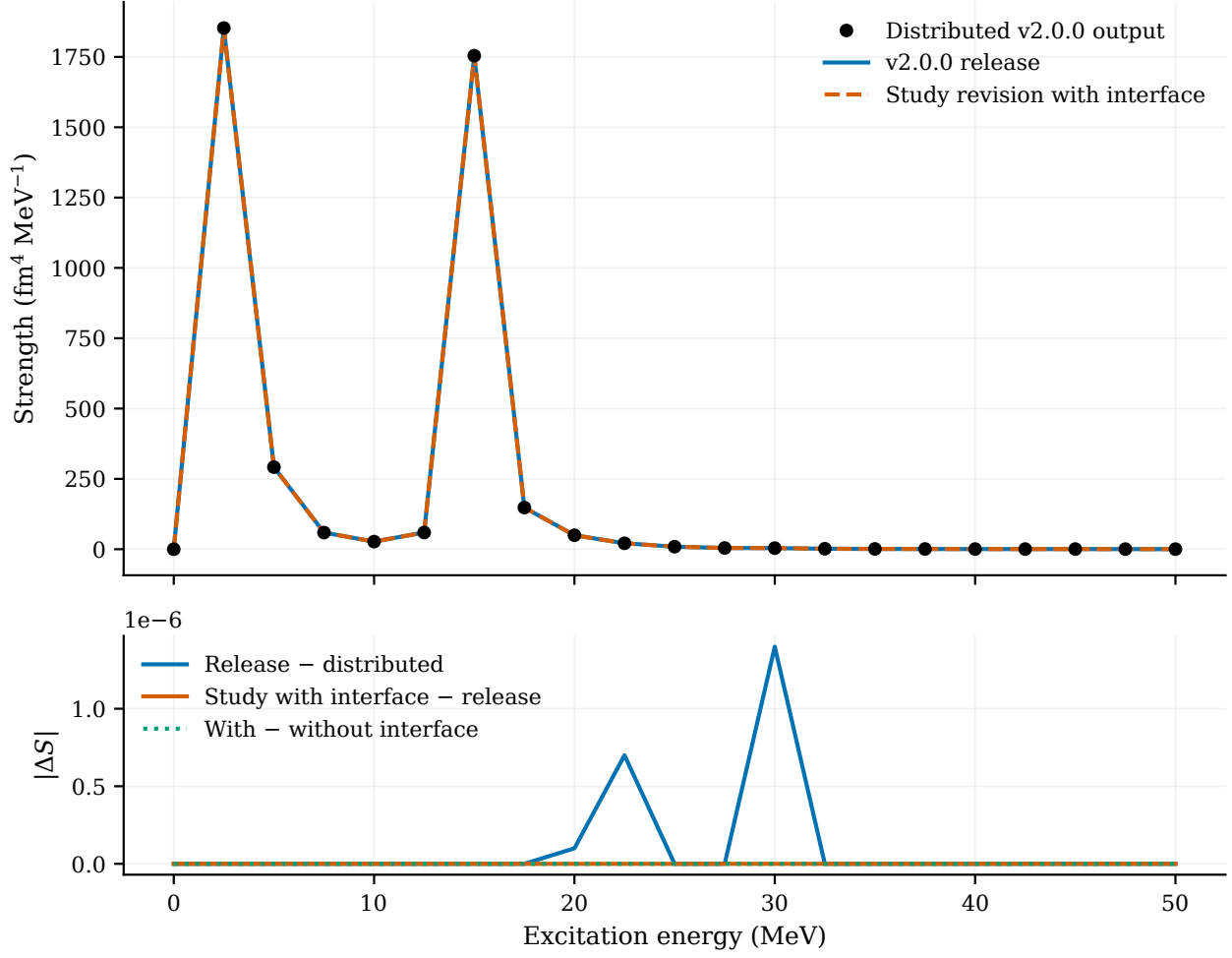


FIG. 6. Official paired ^{152}Sm backend check. The top panel compares the distributed v2.0.0 strength, a release run and the study executable with its interface. The lower panel shows absolute strength differences. No reduced space or PMM is used in this validation.

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